Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is i^2? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Simplify $\sqrt{(-9)}$. Student's answer: ± 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is i^3? Student's answer: i

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Add $(2 + 3i) + (4 - i)$. Student's answer: $6 + 4i$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(3 + i)(3 - i)$. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Multiply $(1 + 2i)(3 + i)$. Student's answer: $5 + 7i$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 + 4 = 0$. Student's answer: $x = \pm 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 + 2x + 5 = 0$. Student's answer: $x = -1 \pm 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x^2 - 6x + 10 = 0$. Student's answer: $x = 3 \pm 2i$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Rewrite $x^2 + 6x + 1$ by completing the square. Student writes: $(x + 3)^2 - 9$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the maximum value of $f(x) = -2x^2 + 8x$? Student's answer: It has a minimum of 8.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the vertex of $y = x^2 - 4x + 7$? Student's answer: $(-2, 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $p(x) = x^2 + 3x - 4$. What is $p(1)$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the remainder when $x^3 - 2x + 1$ is divided by $x - 2$? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $x - 3$ a factor of $x^2 - 5x + 6$? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $6/x = 3$. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $x/(x - 2) = 2$. Student's answer: $x = 2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $1/x + \frac{1}{2} = 1$. Student's answer: $x = 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x^2$ and $y = x + 2$. Student's answer: (2, 4) only

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x^2$ and $y = 4$. Student's answer: (2, 4)

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve the system $y = x^2 - 1$ and $y = 3$. Student's answer: $(2, 3)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the x-values where $y = x^2$ and $y = 2x + 3$ intersect. Student's answer: $x = 3$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find where y = 2^x and y = x + 3 intersect. Student says: They intersect at x = 3.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the x-values where $y = x^3$ and $y = x$ intersect. Student's answer: $x = 1$ only

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the average rate of change of $f(x) = x^2$ on $[1, 3]$. Student's answer: 8

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the average rate of change of $f(x) = 3x - 2$ on $[0, 4]$. Student's answer: 12

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the average rate of change of $f(x) = 2^x$ on $[0, 2]$. Student's answer: 3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Where is the vertex of $f(x) = |x - 2| + 1$? Student's answer: $(-2, 1)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the domain of $f(x) = \sqrt{(x + 4)}$? Student's answer: $x \geq 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $f(x) = x + 1$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. What is $f(-2)$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

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Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is 8^x equivalent to 2^(3x)? Student says: No

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A population is modeled by y = 1000(1.05)^t. What is the annual growth rate? Student's answer: 1.05%

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Rewrite 3^(2t) in the form b^t. Student's answer: 6^t

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$f(x) = 2x + 5$ and g has a constant rate of change of 3. Which function has the greater rate of change? Student's answer: f

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Which grows faster eventually: $f(x) = 2^x$ or $g(x) = x^2$? Student's answer: $g(x) = x^2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

$y = 4x + 1$ and $y = x^2 + 3$ are compared. Which has the greater y-intercept? Student's answer: $y = 4x + 1$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $f(x) = x^2$ and $g(x) = f(x) + 3$. What is $g(2)$? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $g(x) = f(x - 2)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Let $g(x) = 2f(x)$ with $f(x) = x^2$. What is $g(3)$? Student's answer: 9

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Find the inverse of $f(x) = 2x + 3$. Student writes: $f^{-1}(x) = (x + 3)/2$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

If $f(x) = x^3$, what is $f^{-1}(8)$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Is $f(x) = x^2$ one-to-one over all real numbers? Student says: Yes

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is $\log_2 8$? Student's answer: 4

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is $\log_{10} 1000$? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Solve $2^x = 32$. Student's answer: $x = 4$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Convert 180° to radians. Student's answer: 2π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

Convert 90° to radians. Student's answer: π/3

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An arc has radius 5 and central angle 2 radians. What is the arc length? Student's answer: 7

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is $\cos(\pi/2)$? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is sin π? Student's answer: 1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the point on the unit circle at 60°? Student's answer: $\left(\sqrt{\frac{3}{2}}, \frac{1}{2}\right)$

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the amplitude of $y = 3\sin(2x)$? Student's answer: 2

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the period of $y = \sin(2x)$? Student's answer: 2π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

What is the period of $y = \cos(x/3)$? Student's answer: 3π

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A model is $y = 2x + 1$. What is the predicted y-value when $x = 5$? Student's answer: 10

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

An actual value is 12 and the predicted value is 11. What is the residual? Student's answer: -1

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A scatter plot has a correlation coefficient of r = 0.95. Is the linear relationship weak or strong?

Student's answer: weak positive

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A taxi cost is C = 25 + 4m, where m is minutes. What does the slope 4 mean? Student's answer: $25 per minute

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

In the same taxi model C = 25 + 4m, what does the intercept 25 mean? Student's answer: the cost per minute

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.

Find the Error

Name: _____ Date: _____

Analyze the student's work below. Identify the error, then solve the problem correctly.

A temperature model is T = 80 - 2h, where h is hours. What does the slope mean? Student's answer: The temperature rises 2 degrees per hour.

1. What did the student do wrong? Why do you think they made this error?

2. Solve the problem correctly. Show all of your work.